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# Assignment Title:

# Predicting Retail Prices Using Machine Learning for Smart Purchase Decisions

## 1.Introduction

Choosing where to buy goods at the best price is a problem that consumers frequently encounter. Due to supply chains, demand swings, and promotions, prices differ between stores. Although basic comparison tools display current costs, they are unable to forecast future prices or generalize based on inadequate data.

To enable a system that suggests the cheapest store based on discovered trends, this lesson introduces a machine learning method for forecasting product prices across retailers. The approach is presented as a supervised regression issue, and various models are investigated to show variations in performance.

## 2. Learning Objectives

By the end of this tutorial, the reader will:

- Recognize regression-based machine learning for forecasting prices.
- Acquire knowledge of preprocessing structured retail data.
- Examine several machine learning models.
- Use evaluation measures to interpret model performance.
- Construct a system that suggests the least expensive retailer.

## 3. Problem Definition

The assignment is presented as a challenge for supervised learning:

- Inputs: store, product, and time (day)
- Output: predicted price

Predicting prices for each retailer and choosing the one with the lowest estimated price are the objectives. This method converts predictions from machine learning into judgments that can be put into practice.

## 4. Dataset

Realistic pricing conditions are simulated using a synthetic dataset. It consists of:

• Several retailers using various price tactics
• Categories of products with starting prices
• Variation dependent on time to represent seasonal patterns
• Random noise to simulate uncertainty in the actual world

Reproducibility is ensured and reliance on private datasets is avoided by using synthetic data.

## 5. Preprocessing Data

The data needs to be ready before training:

• Numerical encoding is used for categorical variables (store and product).

• Relevance is taken into consideration when choosing features.

• Training and testing sets are separated from the dataset.

These procedures guarantee that performance can be fairly assessed and that the data is appropriate for machine learning models.

## 6. Machine Learning Models

This lesson uses three distinct regression models to forecast housing costs. The complexity and learning capacity of these models differ, making it possible to compare straightforward and sophisticated methods in a meaningful way.

### Linear Regression

One of the most basic machine learning algorithms is linear regression. It is predicated on the idea that the goal variable (home price) and the input attributes have a linear relationship. In order to minimize the discrepancy between expected and actual values, the model tries to fit a straight line (or hyperplane in higher dimensions).

The relationship is represented mathematically as a weighted sum of the input features, each of which makes a separate contribution to the final prediction.

### Random Forest

Several decision trees are combined in the Random Forest ensemble learning technique to provide predictions that are more reliable and accurate.

A random subset of the features and data is used to train each decision tree. The average of each tree's output yields the final prediction. In comparison to a single decision tree, this procedure lessens overfitting and enhances generalization.

Furthermore, Random Forest offers helpful data like feature importance, which enables us to determine which factors have the biggest impact on predictions.

### Gradient Boosting

A more sophisticated ensemble method called gradient boosting develops models consecutively as opposed to independently.

Gradient Boosting trains each new model to fix the mistakes caused by the prior ones, as opposed to training trees in simultaneously (like Random Forest). By concentrating on the residual's differences between expected and actual values this is accomplished.

The model progressively gets better over several iterations by lowering prediction errors.

Among Gradient Boosting's main advantages are:

1. High accuracy of prediction
2. Capacity to identify intricate patterns in data
3. Excellent results on tabular and structured datasets

## 7. Evaluation Metrics

Two metrics are used:

### Mean Absolute Error (MAE)

Measures average prediction error:

$$MAE = \frac{1}{n}\sum | y - \hat{y} |$$

### Root Mean Squared Error (RMSE)

Penalizes large errors more strongly:

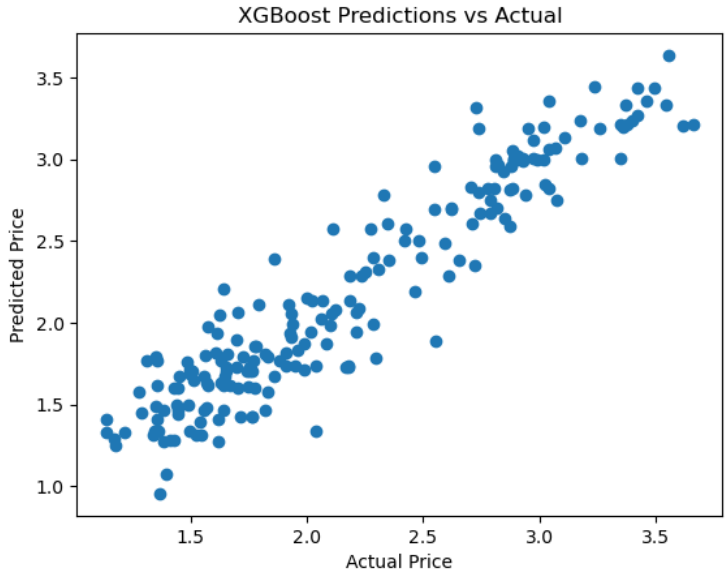
$$RMSE = \sqrt{\frac{1}{n}\sum(y - \hat{y})^2}$$

Better performance is shown by lower values; these metrics give a clear picture of the accuracy and dependability of the model.

# 8. Results and Visualizations

## 8.1 Predicted vs Actual Prices

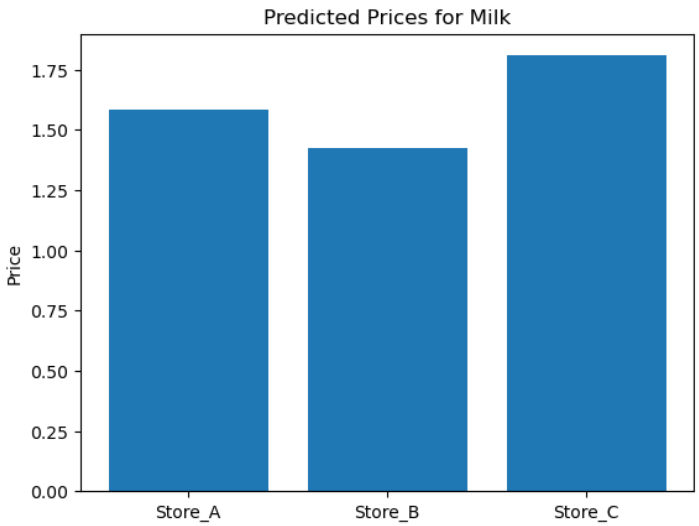
A scatter plot compares predicted and actual values; it indicates how closely predictions match actual values.




This figure shows predicted vs actual prices. Points close to the diagonal indicate accurate predictions.

## 8.2 Price Comparison Across Stores

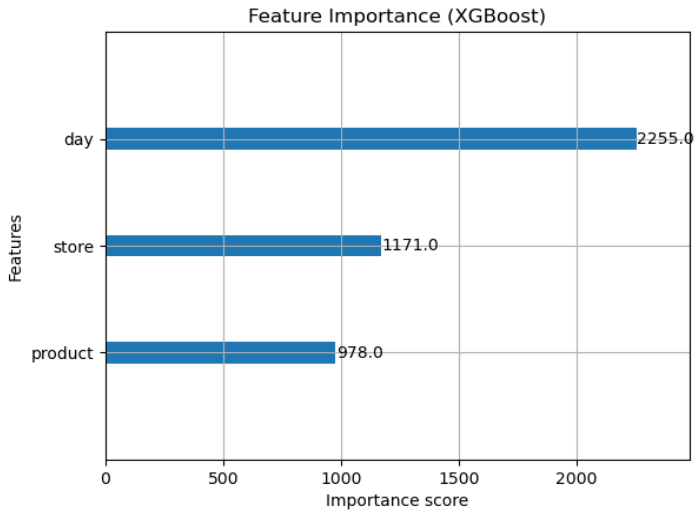
Bar charts show predicted prices for a product across different stores, enabling easy comparison.



### 8.3 Feature Importance

Plots of feature importance show that:

- The store has the biggest impact.
- Pricing is also influenced by product type.
- Seasonal fluctuation is influenced by time.




### 8.4 Model Comparison

Gradient Boosting demonstrated its capacity to simulate intricate relationships by achieving the lowest error.

## 9. Recommendation System

The system chooses the least expensive store based on anticipated prices:

1. Estimate each store's price
2. Examine forecasts
3. Choose the lowest value

Customers receive a useful product as a result.

## 10. Ethical Considerations

Several ethical issues are raised by this system:

• **Bias**: Some stores may be misrepresented due to incomplete data.

• **Transparency**: Users need to be aware that forecasts are approximations.

• **Market impact:** These systems may have an impact on pricing tactics. Fairness and transparency are necessary for responsible use.

## 11. Accessibility Considerations

In-order to guarantee accessibility:

• Colorblind-friendly color schemes are employed, such as blue and orange.
• Graphs have captions and labels that are clear.
• Textual explanations accompany the visuals.
• Color is not the only way to express message.
These actions guarantee inclusion for a wide range of users.

## 12. Conclusion

This tutorial demonstrates the use of machine learning to a practical problem. We create a platform that combines regression models with a recommendation engine to help customers make informed purchases.
The project highlights the importance of data pretreatment, model selection, evaluation, and visualization. It also emphasizes the need of ethical responsibility and accessibility in machine learning applications.

The project emphasizes the significance of:

- Model selection
- Data preprocessing
- Visualization
- Ethical awareness

## 13.References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Pedregosa, F. et al. (2011) *Scikit-learn: Machine learning in Python*. *Journal of Machine Learning Research*, 12, pp. 2825–2830.

Géron, A. (2019) *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.

Scikit-learn (2024) *Regression models documentation*. Available at: https://scikit-learn.org (Accessed: 27 March 2026).


## 14.GitHub Link:

https://github.com/pm25aat/price-prediction-ml